

Chapter1 : MULTITHREADING

Objectives

- Thread properties: thread's name, priority and group
- Create a thread by subclassing the **Thread** class
- Synchronization
- Animation

Section 1: Normal

1. **Thread properties.** Write program to display the properties of current thread, and display numbers in one-second interval.
2. **Two threads.** The following program demonstrates 2 threads which are concurrently running.

Here is the output of running this program:

```
D:\Java>java C
0 A 1 2 B 3 4 C 5 6 7 D 8 9 E F G H I J D:\Java>
```

3. **Synchronization.** The following program demonstrates the use of synchronized keyword. This program displays several strings in the form:

```
<string1>
<string2>
<string3>
```

The program will print "<string1", then it sleeps for a second and then prints ">".

Here is the output of running this program:

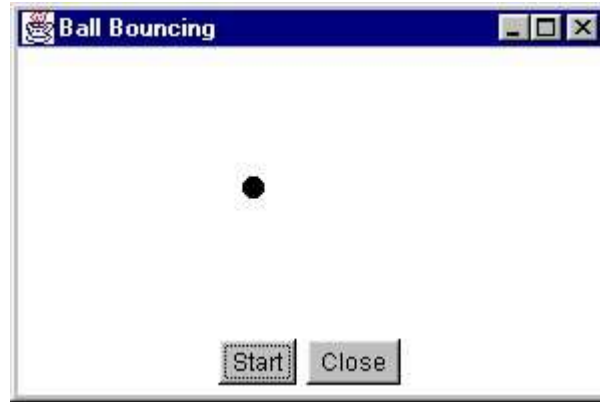
```
C:\java>java C
<Hello>
<Java>
<World>
C:\java>javac C.java
```

Note that without the synchronized keyword, the output will be something like this:

```
C:\java>java C
<Hello<Java<World>
>
>
C:\java>
```

Section 2: Do More (*)

1. **Ball bouncing.** Write program to demonstrate a ball bouncing inside a rectangle as in figure below:



2. Modify sample program 2 so that it has another third thread that displays random strings.
3. Modify sample program 4 so that it has 2 buttons Start and Stop. When you click on Start button, it begins to animate, and when you click on Stop button, it will stop the animation.
4. Modify sample program 5 BallBounce.java to add two more buttons Stop and Continue. When you click on button Stop, the ball will stop moving. When you click on button Continue, the ball continues to move.

Section 3: Do Yourself (**)

1. Write application to display simultaneously random lines on two areas using multithreading.
2. Write application containing three buttons (namely Circle, Line and Both) and one drawing area. Clicking on the Circle button will make the applet to draw random circles, clicking on the Line button will make the applet to draw random lines, and clicking on the Both button will make the applet to draw random circles and lines.
3. Write program to demonstrate transfer of bank accounts. Use thread and synchronization so that there is no account to be unsafe during transactions.

Chapter 2: Remote Method Invocation

Section 1: Normal

1. Write a program depicting remote interface.
2. Write a program that displays random number.
3. Write a program calculating $n!$ using Factory Object
4. Write a program calculating $\text{sum}(n)$ which n key in from keyboard using Factory Object

Section 2: Do more (*)

1. Write the Program that performing Add, sub, Multi, division by RMI
2. Write the program that Matrix Add calculating by RMI
3. Write the Program that performs Reverse Number by clients using Factory Object
4. Write the Program that performs calculate the duplicate string from the file input.dat using Factory Object

Chapter 3: Enterprise JavaBean

Section 1: Normal

1. Develop a bean that has a juggler on the screen. There should be two buttons Start and Stop that starts and stops the juggling.

Modify the above program and increase the juggling speed and then reduce the juggling speed. Save the bean and make an applet.

2. Write a code for a bean representing a simple gauge bean.
3. Write a bean representing the time generator. This bean displays the time zone

Section 2: Do more (*)

1. Write the bean that manages hour working of 10 employees
2. Write the bean that calculates salary total in the month of 10 employees by formula

$$\text{Salary/day} = \text{hour_total} * 12 + \text{OverTimer} * 24$$

Chapter 4: Web Services

Section 1: Normal

1. Write a program that displays random number.
2. Write a program calculating $n!$
3. Write a program calculating $\text{sum}(n)$ which n key in from Keyboard